

$$z^{-1} = e^{-Ts}$$

$$E(z) = Z[e^*(t)] = E^*(s) \Big|_{z=e^{Ts}} = \sum_{n=0}^{\infty} e(nT) \cdot z^{-n}$$

7.4.2 Methods of z-Transform

- By the definition.
- Partial fraction expansion.

7.4.3 Properties of z-Transform

1. linear property $Z[a \cdot e_1^*(t) \pm b \cdot e_2^*(t)] = a \cdot E_1(z) \pm b \cdot E_2(z)$

2. Real shifting theorem 实位移定理

① Lag 延时定理 $Z[e(t - nT)] = z^{-n} E(z)$

Proof. LHS = $\sum_{K=0}^{\infty} e(kT - nT) \cdot z^{-k}$

$\downarrow$ $j = k - n$

$= \sum_{j=-n}^{\infty} e(jT) \cdot z^{-(j+n)} = z^{-n} \sum_{j=0}^{\infty} e(jT) \cdot z^{-j}$

$= z^{-n} E(z) = \text{RHS}$

Left Hand Side (LHS); Right Hand Side (RHS)

2. Real shifting theorem 实位移定理

② Lead 超前定理

$$Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right]$$

Proof.

$$\begin{aligned} \text{LHS} &= \sum_{k=0}^{\infty} e(kT + nT) \cdot z^{-k} = z^n \sum_{k=0}^{\infty} e(kT + nT) \cdot z^{-(k+n)} \\ &\quad \downarrow j = k + n \\ &= z^n \sum_{j=n}^{\infty} e(jT) \cdot z^{-j} = z^n \left[\sum_{j=0}^{\infty} e(jT) \cdot z^{-j} - \sum_{j=0}^{n-1} e(jT) \cdot z^{-j} \right] \\ &= z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right] = \text{RHS} \end{aligned}$$

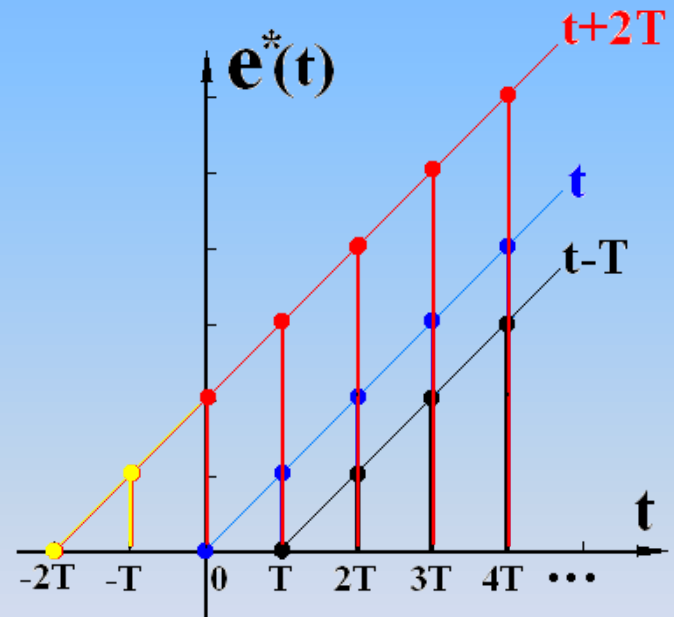
$$Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right]$$

Example 5 $e(t) = t - T$

$$E(z) = Z[t - T] = z^{-1} Z[t] = z^{-1} \frac{Tz}{(z-1)^2} = \frac{T}{(z-1)^2}$$

Example 6 $e(t) = t + 2T$

$$\begin{aligned} E(z) &= Z[t + 2T] \\ &= z^2 \left\{ Z[t] - \sum_{k=0}^1 kT \cdot z^{-k} \right\} \\ &= z^2 \left[\frac{Tz}{(z-1)^2} - 0 - Tz^{-1} \right] \end{aligned}$$



3. Complex shifting theorem 复位移定理

$$Z[e(t) \cdot e^{\mp at}] = E(z \cdot e^{\pm aT})$$

Proof.

$$\text{LHS} = \sum_{k=0}^{\infty} e(kT) \cdot e^{\mp akT} z^{-k} = \sum_{k=0}^{\infty} e(kT) \cdot (z \cdot e^{\pm aT})^{-k}$$

$$\downarrow \quad z_1 = z \cdot e^{\pm aT}$$

$$= \sum_{k=0}^{\infty} e(kT) \cdot (z \cdot e^{\pm aT})^{-k} = E(z_1) = E(z \cdot e^{\pm akT}) = \text{RHS}$$

Example 7 $e(t) = t \cdot e^{-at}$

$$E(z_1) = Z[t]_{z_1 = z \cdot e^{aT}} = \frac{Tz_1}{(z_1 - 1)^2} = \frac{T(z \cdot e^{aT})}{(z \cdot e^{aT} - 1)^2} = \frac{Tz \cdot e^{-aT}}{(z - e^{-aT})^2}$$

4. Initial-value Theorem

$$\lim_{n \rightarrow 0} e(nT) = \lim_{z \rightarrow \infty} E(z)$$

Proof:

$$\begin{aligned} E(z) &= \sum_{n=0}^{\infty} e(nT) \cdot z^{-n} \\ &= [e(0) + e(1) \cdot z^{-1} + e(2) \cdot z^{-2} + e(3) \cdot z^{-3} + \dots] \end{aligned}$$

$$\lim_{z \rightarrow \infty} E(z) = e(0)$$

Example 8
$$E(z) = \frac{0.792 \cdot z^2}{(z-1)[z^2 - 0.416z + 0.208]}$$

$$e(0) = \lim_{z \rightarrow \infty} E(z) = 0$$

Properties of z-Transform

1. linear property $Z[a \cdot e_1^*(t) \pm b \cdot e_2^*(t)] = a \cdot E_1(z) \pm b \cdot E_2(z)$
2. Real shifting theorem
$$\begin{cases} \text{Lag} & Z[e(t - nT)] = z^{-n} E(z) \\ \text{Lead} & Z[e(t + nT)] = z^n \left[E(z) - \sum_{k=0}^{n-1} e(kT) \cdot z^{-k} \right] \end{cases}$$
3. Complex shifting theorem $Z[e(t) \cdot e^{\mp at}] = E(z \cdot e^{\pm aT})$
4. Initial-value theorem $\lim_{n \rightarrow 0} e(nT) = \lim_{z \rightarrow \infty} E(z)$
5. Final-value theorem $\lim_{n \rightarrow \infty} e(nT) = \lim_{z \rightarrow 1} (z - 1) \cdot E(z)$
6. Convolution theorem $c^*(t) = e^*(t) * g^*(t) \Rightarrow C(z) = E(z) \cdot G(z)$

7.4.4 Inverse z-Transform

$$Z^{-1}[X(z)] = x(nT)$$

Tips:

Inverse Z-transform can only provide discrete-time signal $x^*(t)$, instead of continuous signal $x(t)$ 。

{	Long Division (长除法)	Expansion of $\frac{E(z)}{z}$
	Partial-Fraction expansion	
	Residue (留数法)	

$$e(nT) = \sum \text{Res} \left[E(z) \cdot z^{n-1} \right]$$

1. Long Division (长除法)

$$E(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_m}{a_0 z^n + a_1 z^{n-1} + \cdots + a_n}$$

Numerator is divided by denominator ,we get

$$E(z) = c_0 + c_1 z^{-1} + \cdots + c_k z^{-k} + \cdots = \sum_{k=0}^{\infty} c_k z^{-k} = \sum_{k=0}^{\infty} e(kT) z^{-k}$$

$$e^*(t) = c_0 \delta(t) + c_1 \delta(t-T) + \cdots + c_k \delta(t-kT) + \cdots$$

Example: $E(z) = \frac{10z}{(z-1)(z-2)}$, obtain $e^*(t)$.

Solution: $E(z) = \frac{10z}{z^2 - 3z + 2} = 10z^{-1} + 30z^{-2} + 70z^{-3} + 150z^{-4} + \dots$

$$z^2 - 3z + 2 \overline{) \begin{array}{l} 10z^{-1} + 30z^{-2} + 70z^{-3} + 150z^{-4} \dots \\ 10z \end{array}}$$

$$\underline{10z - 30z^0 + 20z^{-1}}$$

$$30z^0 - 20z^{-1}$$

$$\underline{30z^0 - 90z^{-1} + 60z^{-2}}$$

$$70z^{-1} - 60z^{-2}$$

$$\underline{70z^{-1} - 210z^{-2} + 140z^{-3}}$$

$$150z^{-2} - 140z^{-3}$$

...

$$\begin{aligned} e^*(t) = & 10\delta(t-T) \\ & + 30\delta(t-2T) \\ & + 70\delta(t-3T) \\ & + 150\delta(t-4T) \\ & + \dots \end{aligned}$$

Example

$$F(z) = \frac{z}{(z-2)(z-3)}, \text{ obtain } f^*(t).$$

Solution:

$$\text{Because } F(z) = \frac{z}{z^2 - 5z + 6} = \frac{z^{-1}}{1 - 5z^{-1} + 6z^{-2}}$$

By long-division, we get that

$$F(z) = z^{-1} + 5z^{-2} + 19z^{-3} + 65z^{-4} + \dots$$

Thus

$$f(0) = 0, \quad f(T) = 1, \quad f(2T) = 5, \quad f(3T) = 19, \quad f(4T) = 65, \dots$$


Then

$$f^*(t) = \delta(t-T) + 5\delta(t-2T) + 19\delta(t-3T) + 65\delta(t-4T) + \dots$$

2. Partial fraction expansion

Note: here, we expand $\frac{X(z)}{z}$, instead of $X(z)$.

$$\frac{X(z)}{z} = \sum_{i=1}^n \frac{A_i}{z - z_i}$$

Consider

$$X(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_{m-1} z + b_m}{a_0 z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n}$$

Then

$$X(z) = \frac{b_0 z^m + b_1 z^{m-1} + \cdots + b_{m-1} z + b_m}{a_0 \prod_{i=1}^n (z - z_i)}$$

If there is no repeated root for the denominator, it generates

$$X(z) = z \left(\frac{A_1}{z - z_1} + \frac{A_2}{z - z_2} + \cdots + \frac{A_n}{z - z_n} \right)$$

其中系数 A_i , 可由式决定:

$$A_i = \left[(z - z_i) \frac{X(z)}{z} \right] \Big|_{z=z_i}$$

Example Consider

$$F(z) = \frac{z}{(z-1)(z-e^{-T})}$$

Obtain $f^*(t)$.

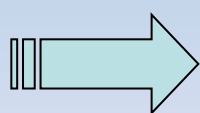
Solution:

$$\frac{F(z)}{z} = \frac{K_1}{z-1} + \frac{K_2}{z-e^{-T}}$$

$$K_1 = \lim_{z \rightarrow 1} \left(\frac{z-1}{z} \right) F(z) = \frac{1}{1-e^{-T}}$$

$$F(z) = \frac{1}{1-e^{-T}} \left(\frac{z}{z-1} - \frac{z}{z-e^{-T}} \right)$$

$$K_2 = \lim_{z \rightarrow e^{-T}} \left(\frac{z-e^{-T}}{z} \right) F(z) = -\frac{1}{1-e^{-T}}$$



$$f(nT) = \frac{1}{1-e^{-T}} (1 - e^{-nT})$$

$$f^*(t) = \frac{1}{1-e^{-T}} \sum_{k=0}^{+\infty} (1 - e^{-kT}) \delta(t - kT)$$

Example 11 $E(z) = \frac{z^2}{(z-0.8)(z-0.1)}$ Obtain $e^*(t)$. (PFE)

PFE:

$$\frac{E(z)}{z} = \frac{z}{(z-0.8)(z-0.1)} = \frac{C_1}{(z-0.8)} + \frac{C_2}{(z-0.1)}$$

$$C_1 = \lim_{z \rightarrow 0.8} \frac{z}{(z-0.1)} = \frac{8}{7} \quad C_2 = \lim_{z \rightarrow 0.1} \frac{z}{(z-0.8)} = \frac{-1}{7}$$

$$= \frac{8/7}{(z-0.8)} - \frac{1/7}{(z-0.1)}$$

$$E(z) = \frac{8}{7} \cdot \frac{z}{(z-0.8)} - \frac{1}{7} \cdot \frac{z}{(z-0.1)}$$

$$e(t) = (8 \times 0.8^{\frac{t}{T}} - 0.1^{\frac{t}{T}}) / 7 \quad e(nT) = (8 \times 0.8^n - 0.1^n) / 7$$

$$e^*(t) = \sum_{n=0}^{\infty} [(8 \times 0.8^n - 0.1^n) / 7] \cdot \delta(t - nT)$$

3、Residue(留数法)

$$F(z) = \sum_{k=0}^{+\infty} f(kT)z^{-k}$$

$$F(z)z^{m-1} = \sum_{k=0}^{+\infty} f(kT)z^{m-k-1}$$

Γ Encircle all the poles of $F(z)z^{k-1}$

$$\oint_{\Gamma} F(z)z^{m-1}dz = \oint_{\Gamma} \left[\sum_{k=0}^{+\infty} f(kT)z^{m-k-1} \right] dz$$

$$\oint_{\Gamma} F(z)z^{m-1}dz = \sum_{k=0}^{+\infty} f(kT) \oint_{\Gamma} z^{m-k-1} dz$$

When $m=k$,

$$f(kT) = \sum_{i=1}^n \text{res}[F(z)z^{k-1}, z_i]$$

$z_i, i = 1, 2, \dots, n$ are all the poles of $F(z)z^{k-1}$

$$\text{Res}\left[z^{(k-1)}x(z)\right]=\lim_{z\rightarrow z_i}\frac{1}{(r-1)!}\frac{d^{r-1}}{dz^{r-1}}\left[(z-z_i)^r z^{k-1}x(z)\right]$$

其中 $\text{Res}[\]$ 表示函数的留数, r 为极点的阶数。

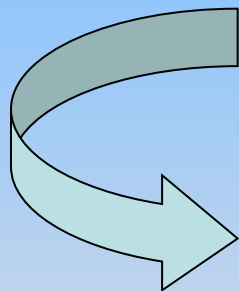
Example For

$$F(z)=\frac{10z}{(z-1)(z-2)}$$

Obtain its inverse z-transform by residue Method.

Solution: $F(z)z^{k-1} = \frac{10z^k}{(z-1)(z-2)}$

Poles $z_1 = 1$ and $z_2 = 2$, and


$$\begin{aligned} \text{res}[F(z)z^{k-1}, 1] &= \lim_{z \rightarrow 1} (z-1)F(z)z^{k-1} = -10 \\ \text{res}[F(z)z^{k-1}, 2] &= \lim_{z \rightarrow 2} (z-2)F(z)z^{k-1} = 10 \cdot 2^k \end{aligned}$$

Then $f(kT) = 10(2^k - 1) \quad (k = 0, 1, 2, \dots)$

Example 12 $E(z) = \frac{5}{(z-a)^2}$ Obtain $e^*(t)$. (Residue)

Solution.

$$e(nT) = \sum \text{Res} \left[E(z) \cdot z^{n-1} \right] = \text{Res}_{z=a} \left[\frac{5}{(z-a)^2} \cdot z^{n-1} \right]$$

$$e(nT) = \frac{1}{(2-1)!} \lim_{z \rightarrow a} \frac{d}{dz} \left[(z-a)^2 \frac{5 \cdot z^{n-1}}{(z-a)^2} \right]$$

$$= \lim_{z \rightarrow a} \frac{d}{dz} [5 \cdot z^{n-1}]$$

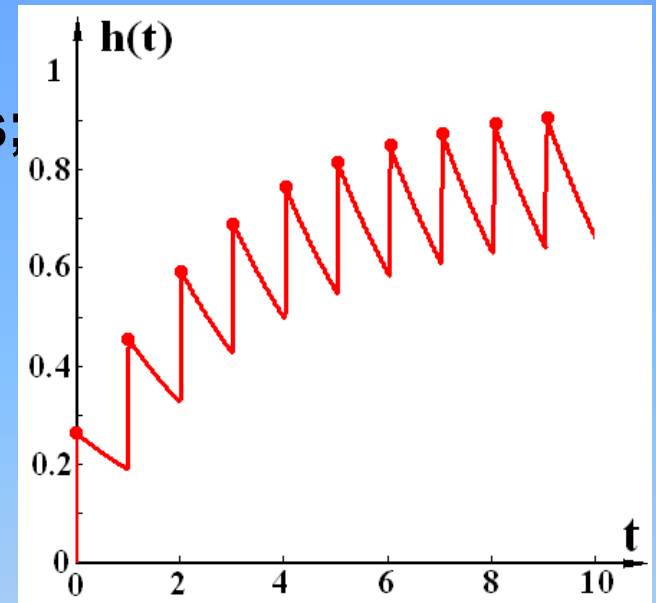
$$= 5 \cdot \lim_{z \rightarrow a} [(n-1) \cdot z^{n-2}]$$

$$= 5 \cdot (n-1) \cdot a^{n-2}$$

$$e^*(t) = \sum_{n=0}^{\infty} (5(n-1) \cdot a^{n-2}) \cdot \delta(t - nT)$$

7.4.5 Limitations of z-Transform

- (1) only shows the information of samples;
- (2) In some cases, the continuous signal may jump on the sampling point.



Chapter 7 Analysis and Design of Linear Discrete-Time System (Sampled-data System)

7.1 Introduction

7.2 The Sampling Process and Sampling Theorem

7.3 Signal Recovery and Zero-Order Hold

7.4 Z-Transform and Inverse Z Transform

7.5 Mathematical Models of Discrete-Time Systems

7.6 Performance Analysis of Discrete-Time Systems

7.7 Digital Control Design for Discrete-Time Systems

7.5 Mathematical Models of Discrete-Time Systems

- **Difference Equation**
- **Impulse Transfer function**

7.5.1 Linear Time-Invariant Difference Equations

(1) Definition of difference $e(kT) = e(k)$

Forward difference	{	First-order	$\Delta e(k) = e(k+1) - e(k)$	$\lim_{T \rightarrow 0} \frac{\Delta e(k)}{T} = \frac{de(t)}{dt}$
		Second-order	$\Delta^2 e(k) = \Delta e(k+1) - \Delta e(k)$	
		:	$= e(k+2) - 2e(k+1) + e(k)$	
		:		
		nth-order	$\Delta^n e(k) = \Delta^{n-1} e(k+1) - \Delta^{n-1} e(k)$	

$$\text{Backward difference} \left\{ \begin{array}{ll} \text{First-order} & \nabla e(k) = e(k) - e(k-1) \\ \text{Second-order} & \nabla^2 e(k) = \nabla e(k) - \nabla e(k-1) \\ & = e(k) - 2e(k-1) + e(k-2) \\ \vdots & \\ \text{nth-order} & \nabla^n e(k) = \nabla^{n-1} e(k) - \nabla^{n-1} e(k-1) \end{array} \right. \quad \lim_{T \rightarrow 0} \frac{\nabla e(k)}{T} = \frac{de(t)}{dt}$$

(2) Difference equation

The equation of the input, output and their higher order differences.

The (forward) difference equation of nth-order linear time-invariant discrete system.

$$\begin{aligned} c(k+n) + a_1c(k+n-1) + a_2c(k+n-2) + \cdots + a_{n-1}c(k+1) + a_nc(k) \\ = b_0r(k+m) + b_1r(k+m-1) + \cdots + b_{m-1}r(k+1) + b_mr(k) \end{aligned}$$

The (backward) differential equation of n-order linear time-invariant discrete system.

$$\begin{aligned} c(k) + a_1c(k-1) + a_2c(k-2) + \cdots + a_{n-1}c(k-n+1) + a_nc(k-n) \\ = b_0r(k-n+m) + b_1r(k-n+m-1) + \\ \cdots + b_{m-1}r(k-n+1) + b_mr(k-n) \end{aligned}$$

(3) To solve difference equations: $\begin{cases} \text{Iteration method} \\ \text{Z-transform method} \end{cases}$

Example 1 The differential equation of a continuous system is:
$$\begin{cases} \ddot{e}(t) - 4\dot{e}(t) + 3e(t) = r(t) = 1(t) \\ e(t) = 0 \quad (t \leq 0) \end{cases}$$

Obtain the corresponding forward difference equation and its solution.

Solution.

$$\dot{e}(t) \approx \frac{\Delta e(k)}{T} = \frac{e(k+1) - e(k)}{T} \stackrel{T=1}{=} e(k+1) - e(k)$$

$$\ddot{e}(t) \approx \frac{\Delta^2 e(k)}{T^2} = \frac{\Delta e(k+1)/T - \Delta e(k)/T}{T} \stackrel{T=1}{=} e(k+2) - 2e(k+1) + e(k)$$

$$\begin{array}{r} e(k+2) - 2e(k+1) + e(k) \\ -4[e(k+1) - e(k)] \\ +3[e(k)] \\ \hline e(k+2) - 6e(k+1) + 8e(k) = 1(k) \end{array}$$

$$\begin{cases} e(k+2) - 6e(k+1) + 8e(k) = 1(k) \\ e(k) = 0 \quad (k \leq 0) \end{cases}$$

Solution I of the difference equation — **Iteration method**

$$\begin{cases} e(k+2) - 6e(k+1) + 8e(k) = 1(k) \\ e(k) = 0 \quad (k \leq 0) \end{cases}$$

Solution $e(k+2) = 6e(k+1) - 8e(k) + 1(k)$

$$k = -1: e(1) = 6e(0) - 8e(-1) + 1(-1) = 0$$

$$k = 0: e(2) = 6e(1) - 8e(0) + 1(0) = 0 - 0 + 1 = 1$$

$$k = 1: e(3) = 6e(2) - 8e(1) + 1(1) = 6 - 0 + 1 = 7$$

$$k = 2: e(4) = 6e(3) - 8e(2) + 1(2) = 6 \times 7 - 8 \times 1 + 1 = 35$$

$\vdots$

$$e^*(t) = \delta(t-2) + 7\delta(t-3) + 35\delta(t-4) + \dots$$

Solution II of difference equation — Z-transform method

$$e(k+2) - 6e(k+1) + 8e(k) = 1(k)$$

$$\begin{aligned} Z : \quad & z^2 [E(z) - e(0)z^0 - e(1)z^{-1}] \\ & - 6 \cdot z [E(z) - e(0)z^0] \\ & + 8 [E(z)] \end{aligned}$$

$$\begin{cases} e(k+2) - 6e(k+1) + 8e(k) = 1(k) \\ e(k) = 0 \quad (k \leq 0) \end{cases}$$

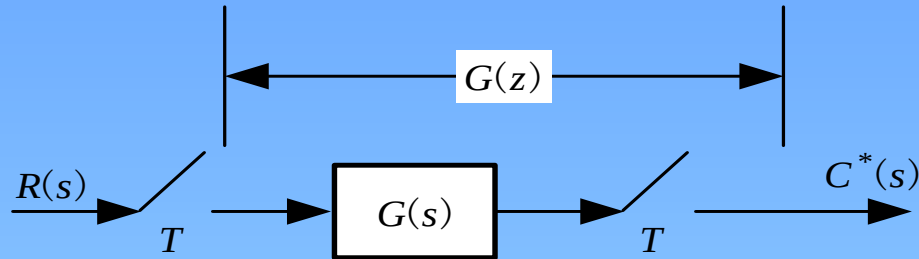
$$\frac{(z^2 - 6z + 8)E(z) = Z[1(k)] = \frac{z}{z-1}}{E(z) = \frac{z}{(z-1)(z-2)(z-4)}}$$

$$Z^{-1} : e(n) = \sum \text{Res} [E(z) \cdot z^{n-1}]$$

$$= \lim_{z \rightarrow 1} \frac{z \cdot z^{n-1}}{(z-2)(z-4)} + \lim_{z \rightarrow 2} \frac{z \cdot z^{n-1}}{(z-1)(z-4)} + \lim_{z \rightarrow 4} \frac{z \cdot z^{n-1}}{(z-1)(z-2)} = \frac{1}{3} - \frac{2^n}{2} + \frac{4^n}{6}$$

$$e^*(t) = \sum_{n=0}^{\infty} e(nT) \cdot \delta(t - nT) = \sum_{n=0}^{\infty} \left(\frac{1}{3} - \frac{2^n}{2} + \frac{4^n}{6} \right) \cdot \delta(t - nT)$$

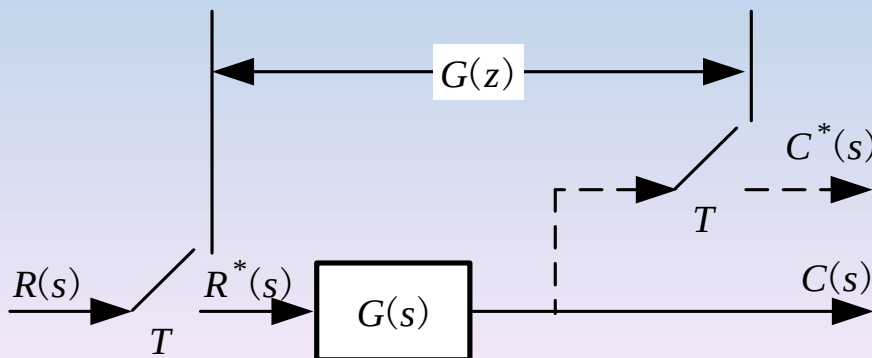
7.5.2 Mathematical Models in Complex Domain — Impulse Transfer Function (脉冲传递函数)

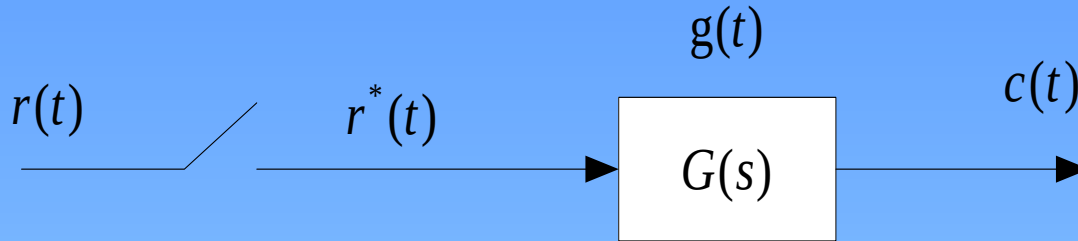


1. Definition

The ratio of the z-T. of the output to the z-T. of the input under zero initial condition.

$$G(z) = \frac{C(z)}{R(z)}$$





$$r^*(t) = \sum_{n=0}^{\infty} r(nT) \delta(t - nT)$$

$$\therefore r^*(t) = r(0)\delta(t) + r(T)\delta(t - T) + \cdots + r(nT)\delta(t - nT) + \cdots$$

$$\therefore c(t) = r(0)g(t) + r(T)g[t - T] + \cdots + r(nT)g[t - nT] + \cdots$$

$$c(kT) = r(0)g(kT) + r(T)g[(k - 1)T] + \cdots + r(nT)g[(k - n)T] + \cdots$$

$$= \sum_{n=0}^{\infty} r(nT)g[(k - n)T]$$

$$c(kT) = \sum_{n=0}^{\infty} r(nT)g[(k-n)T]$$

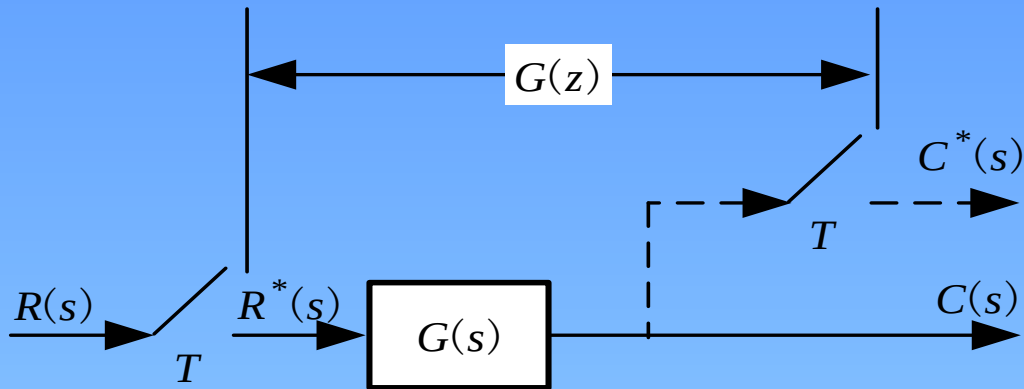
$$C(z) = \sum_{k=0}^{\infty} c(kT)z^{-k} = \sum_{k=0}^{\infty} \sum_{n=0}^{\infty} r(nT)g[(k-n)T]z^{-k}$$

$$= \sum_{n=0}^{\infty} r(nT)z^{-n} \sum_{k=0}^{\infty} g[(k-n)T]z^{-(k-n)}$$

The z-transform of
unity impulse
response sequence

$$\therefore G(z) = \frac{C(z)}{R(z)} = \sum_{k=n}^{\infty} g[(k-n)T]z^{-(k-n)} = \sum_{j=0}^{\infty} g(jT)z^{-j}$$

$$\mathbf{G(Z)=Z[g(t)]=Z[G(s)]}$$



Example 1 Consider the discrete system shown in the figure with

$$G(s) = \frac{1}{s(0.1s + 1)}$$

Obtain the impulse-transfer function $G(z)$.

Solution:

Method I. The impulse response is:

$$g(t) = (1 - e^{-10t}) \quad (t > 0)$$

$$g(kT) = 1 - e^{-10kT}$$

Then the impulse transfer function is:

$$\begin{aligned} G(z) &= \sum_{k=0}^{+\infty} g(kT)z^{-k} = \sum_{k=0}^{+\infty} (1 - e^{-10kT})z^{-k} \\ &= \frac{z}{z-1} - \frac{z}{z-e^{-10T}} = \frac{z(1 - e^{-10T})}{(z-1)(z-e^{-10T})} \end{aligned}$$

Method II. Because $G(s) = \frac{1}{s} - \frac{1}{s+10}$

Then by $G(Z) = Z[g(t)] = Z[G(s)]$, it derives

$$G(z) = \frac{z}{z-1} - \frac{z}{z-e^{-10T}} = \frac{z(1 - e^{-10T})}{(z-1)(z-e^{-10T})}$$

The properties of impulse transfer function:

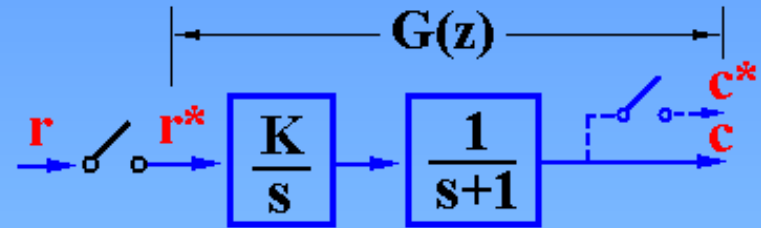
- (1) $G(z)$ is a complex function of complex variable z ;**
- (2) $G(z)$ depends only on the structure and parameters of the system;**
- (3) $G(z)$ has a relation with the difference equation of the system;**
- (4) $G(z)$ is equal to $Z[g^*(t)]$;**
- (5) $G(z) \sim$ zero-pole location in z plane.**

The limitation of impulse-transfer functions

- (1) It can not reflect the full information of the system response under non-zero initial conditions;**
- (2) It is only for SISO discrete systems;**
- (3) It is only for linear time-invariant difference equations;**

Example 2 Consider the discrete system shown in the figure ($T=1$). Obtain

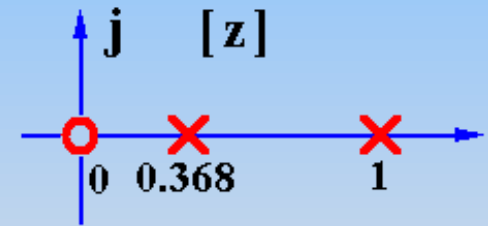
- (1) Impulse-transfer function of the system
- (2) Zero-poles location in z plane;
- (3) Difference equation of the system.



Solution. (1) $G(z) = \frac{C(z)}{R(z)} = Z \left[\frac{K}{s(s+1)} \right] = K \cdot Z \left[\frac{1}{s} - \frac{1}{s+1} \right]$

$$= K \left[\frac{z}{z-1} - \frac{z}{z-e^{-T}} \right] = \frac{(1-e^{-T})Kz}{(z-1)(z-e^{-T})} = \frac{(1-e^{-T})Kz}{z^2 - (1+e^{-T})z + e^{-T}}$$

$$= \frac{0.632Kz^{-1}}{1 - 1.368z^{-1} + 0.368z^{-2}}$$

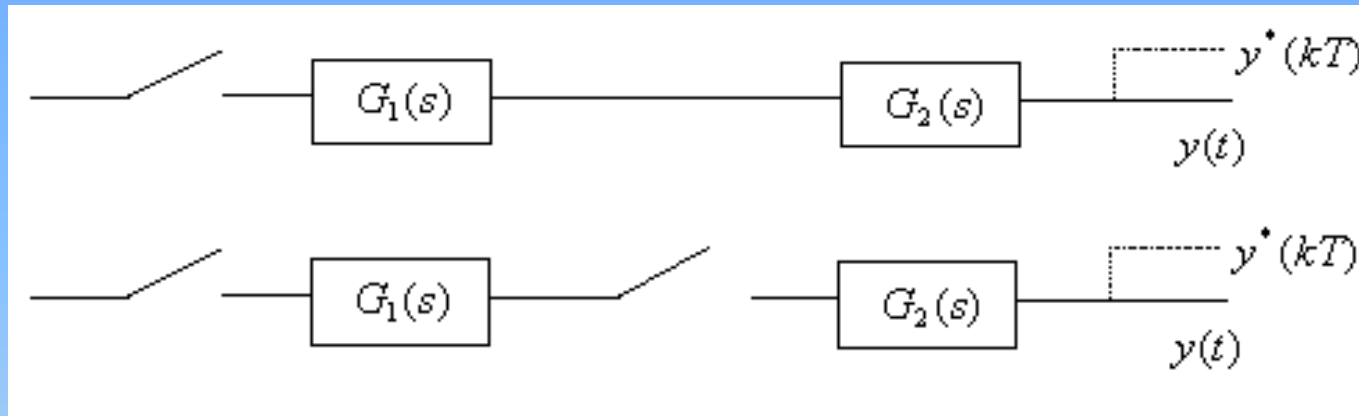


(2) Zero-poles location in z plane

(3) $(1 - 1.368z^{-1} + 0.368z^{-2})C(z) = 0.632Kz^{-1}R(z)$

$$c(k) - 1.368c(k-1) + 0.368c(k-2) = 0.632Kr(k-1)$$

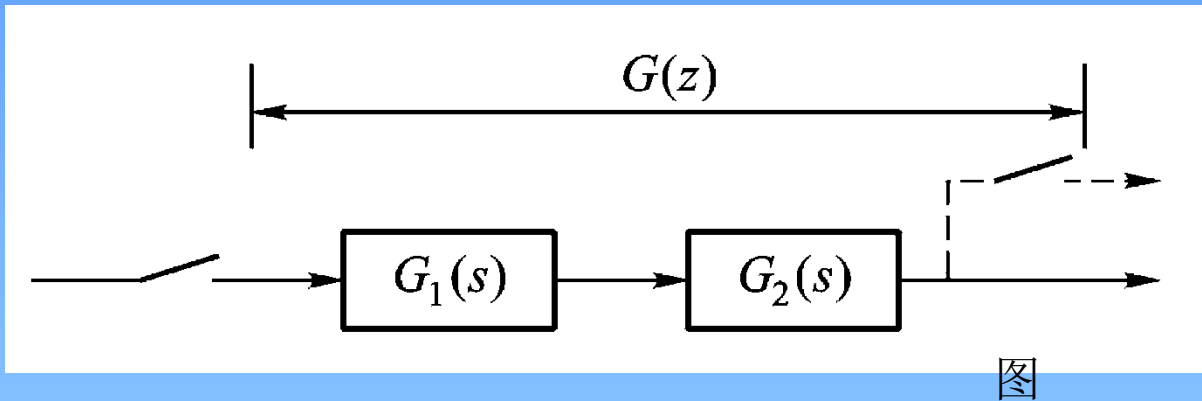
7.5.3 Impulse transfer function of Open-Loop Systems



(1) There is no sampler/switch between two components

$$G(s) = G_1(s)G_2(s)$$

$$G(z) = Z[G_1(s)G_2(s)] = G_1G_2(z)$$



Example 3 Consider the discrete system shown in the above figure , where

$$G_1(s) = \frac{1}{s + a} \qquad G_2(s) = \frac{1}{s + b}$$

Obtain the open-loop impulse transfer function.

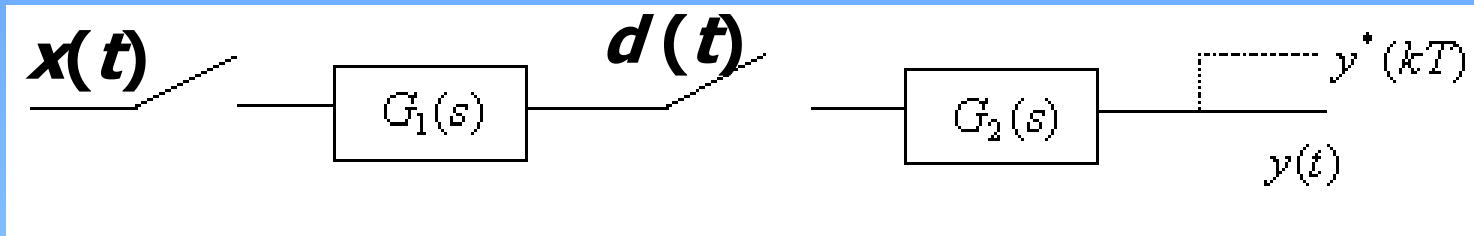
solution:

$$G_1(s)G_2(s) = \frac{1}{b-a} \left[\frac{1}{s+a} - \frac{1}{s+b} \right]$$

$$G(z) = G_1G_2(z)$$

$$= \frac{1}{b-a} \left[\frac{z(e^{-aT} - e^{-bT})}{(z - e^{-aT})(z - e^{-bT})} \right]$$

(2) There is a sampler/switch between two components



$$D(z) = G_1(z)X(z)$$

$$Y(z) = G_2(z)D(z) = G_1(z)G_2(z)R(z)$$

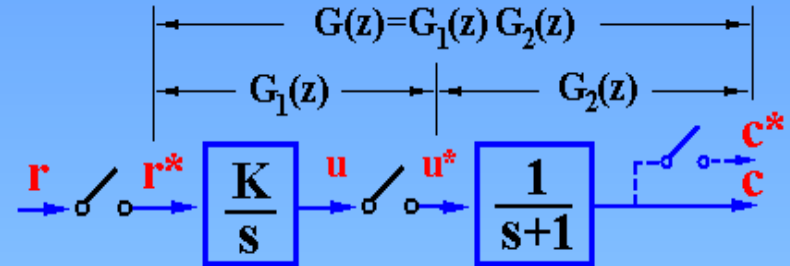
$$\therefore G(z) = G_1(z)G_2(z)$$

注 $G_1(z)G_2(z) \neq G_1G_2(z)$

(1) Switch between factors

$$G(z) = G_1(z) G_2(z) = Z \left[\frac{K}{s} \right] \cdot Z \left[\frac{1}{s+1} \right]$$

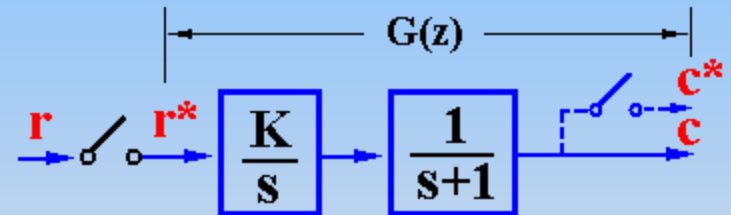
$$= \frac{Kz}{z-1} \cdot \frac{z}{z-e^{-T}} = \frac{Kz^2}{(z-1)(z-e^{-T})}$$



(2) No switch between factors

$$G(z) = Z[G_1(s) \cdot G_2(s)] = G_1 G_2(z)$$

$$= K \left[\frac{z}{z-1} - \frac{z}{z-e^{-T}} \right] = \frac{(1-e^{-T})Kz}{(z-1)(z-e^{-T})}$$



Note: the zeros of $G(z)$, the poles of $G(z)$.

Exercise: Consider $G_1(s) = \frac{1}{s}$, $G_2(s) = \frac{10}{s+10}$, obtain $G(z)$.

Solution:

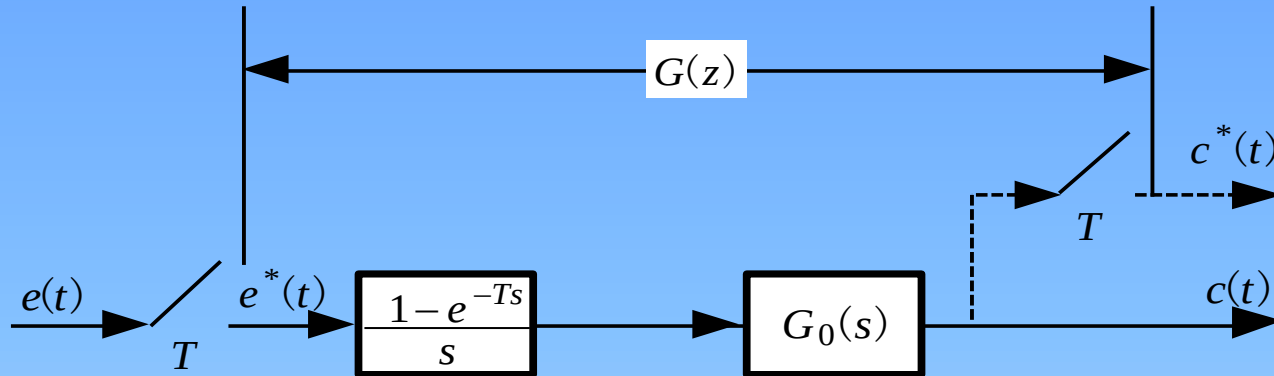
If there is no switch between the components,

$$G(z) = G_1 G_2(z) = Z\left[\frac{10}{s(s+10)}\right] = \frac{z(1-e^{-10T})}{(z-1)(z-e^{-10T})}$$

If there is a sampler between the components,,

$$\begin{aligned} G(z) &= G_1(z)G_2(z) = Z\left[\frac{1}{s}\right]Z\left[\frac{10}{s+10}\right] \\ &= \frac{z}{z-1} \frac{10z}{z-e^{10T}} = \frac{10z^2}{(z-1)(z-e^{-10T})} \end{aligned}$$

(3) ZOH in the system



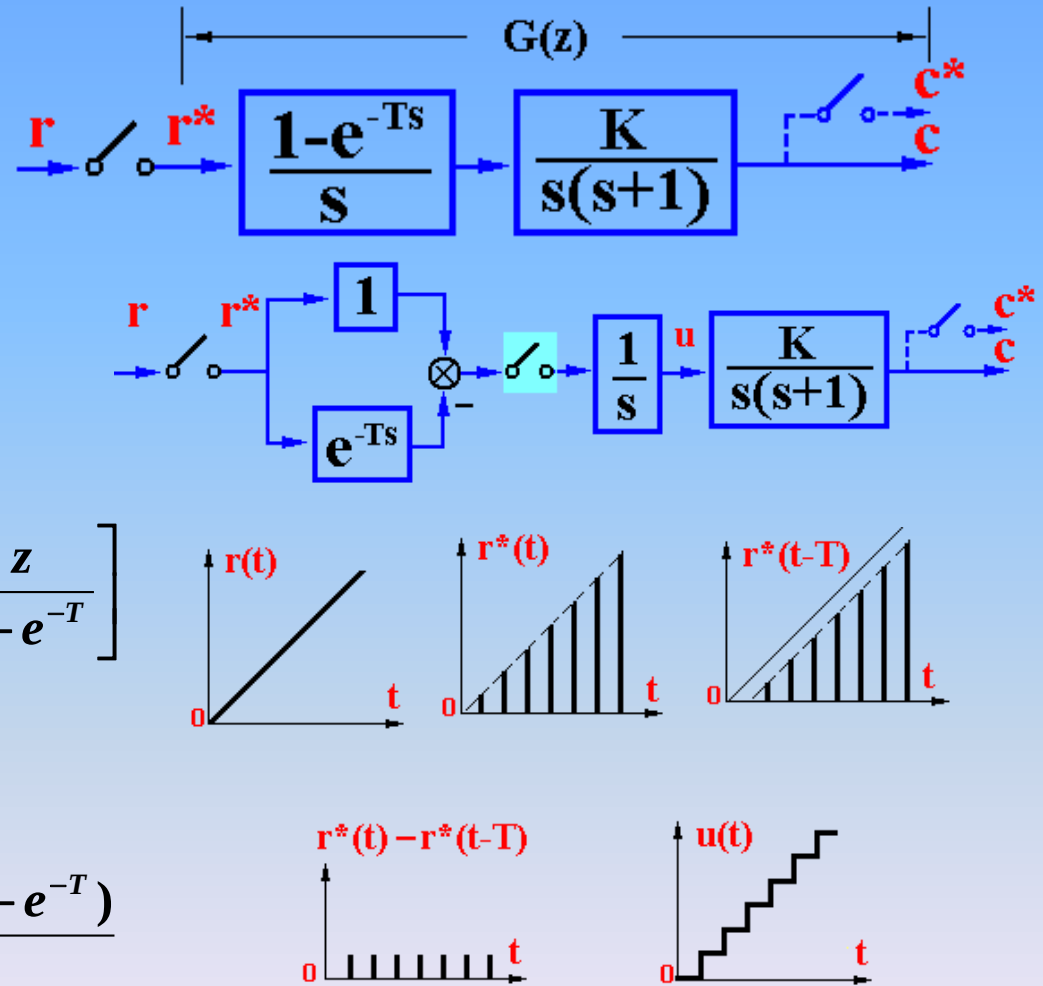
$$C(z) = Z\left[\frac{1 - e^{-Ts}}{s} G_0(s)\right] R(z) = Z\left[\frac{1}{s} G_0(s) - \frac{e^{-Ts}}{s} G_0(s)\right] R(z)$$

$$Z\left[\frac{e^{-Ts}}{s} G_0(s)\right] = z^{-1} Z\left[\frac{G_0(s)}{s}\right] \quad C(z) = (1 - z^{-1}) Z\left[\frac{G_0(s)}{s}\right] R(z)$$

$$G(z) = \frac{C(z)}{R(z)} = (1 - z^{-1}) Z\left[\frac{G_0(s)}{s}\right]$$

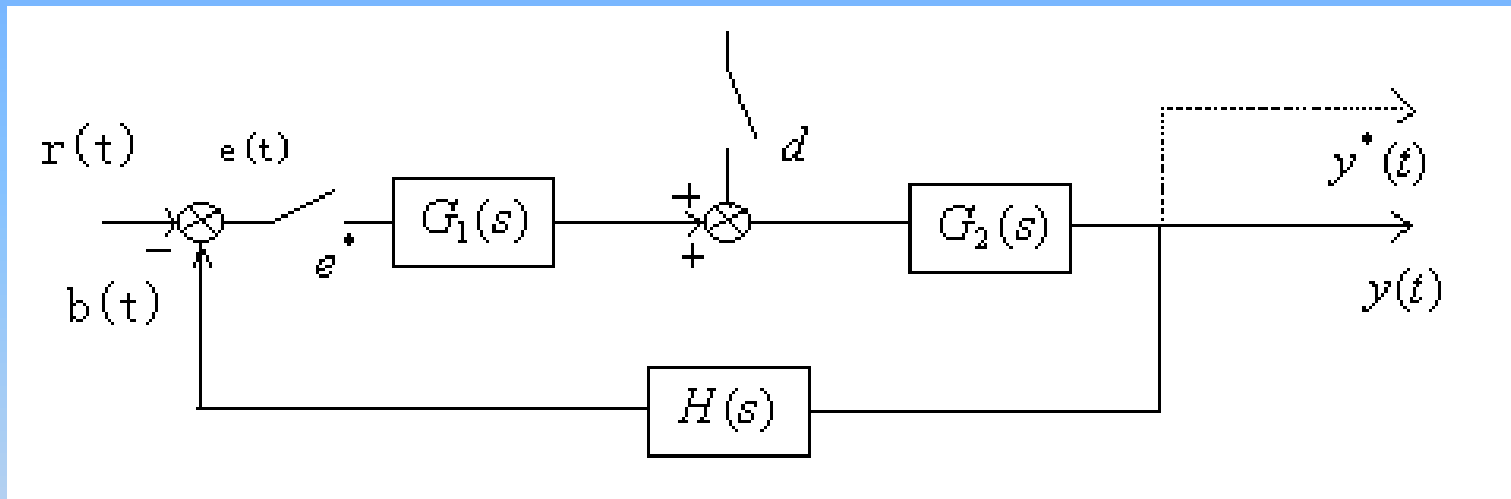
Example 4 Consider the discrete system shown in the following figure, obtain its impulse transfer function.

$$\begin{aligned}
 G(z) &= Z \left[\frac{1-e^{-Ts}}{s} \cdot \frac{K}{s(s+1)} \right] \\
 &= K(1-z^{-1})Z \left[\frac{1}{s^2(s+1)} \right] \\
 &= K \frac{z-1}{z} Z \left[\frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1} \right] \\
 &= K \frac{z-1}{z} \left[\frac{Tz}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-T}} \right] \\
 &= K \left[\frac{T}{z-1} - 1 + \frac{z-1}{z-e^{-T}} \right] \\
 &= K \frac{(T-1+e^{-T})z + (1-Te^{-T}-e^{-T})}{(z-1)(z-e^{-T})}
 \end{aligned}$$



ZOH does not change the system order and O.-L. poles but changes the O.-L. zeros.

7.5.4、 Impulse transfer function of Closed-Loop Systems



(1) 、 Impulse Transfer Function for input to output.

$$d = 0$$

$$Y(z) = G_1 G_2(z) E(z)$$

$$\left. \begin{aligned} e(t) &= r(t) - b(t) \\ \Rightarrow E(z) &= R(z) - B(z) \\ B(z) &= G_1 G_2 H(z) E(z) \end{aligned} \right\} \Rightarrow E(z) = \frac{R(z)}{1 + G_1 G_2 H(z)}$$

Error impulse transfer function (误差脉冲传递函数):

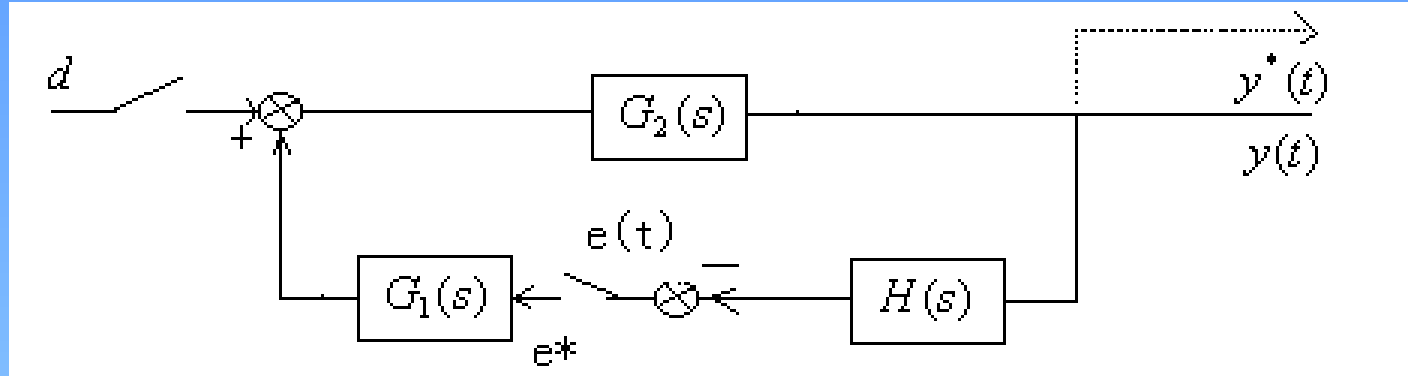
$$G_e(z) = \frac{E(z)}{R(z)} = \frac{1}{1 + G_1 G_2 H(z)}$$

$$\Rightarrow Y(z) = G_1 G_2(z) \frac{R(z)}{1 + G_1 G_2 H(z)}$$

$$\therefore \Phi(z) = \frac{Y(z)}{R(z)} = \frac{G_1 G_2(z)}{1 + G_1 G_2 H(z)}$$

(2) Impulse Transfer Function for disturbance to output

$$r(t) = 0$$



$$Y(z) = G_2(z)D(z) + G_1G_2(z)E(z)$$

$$E(z) = -[G_2H(z)D(z) + G_1G_2H(z)E(z)]$$

$$\Rightarrow E(z) = -\frac{G_2H(z)}{1 + G_1G_2H(z)} D(z)$$

$$\therefore Y(z) = G_2(z)D(z) - \frac{G_1G_2(z)G_2H(z)}{1 + G_1G_2H(z)} D(z)$$

$$\Rightarrow \Phi_d(z) = \frac{Y(z)}{D(z)} = G_2(z) - \frac{G_1G_2(z)G_2H(z)}{1 + G_1G_2H(z)}$$

E(z) :

**D(z) passing through
G₂(z);**

Loop of E(z)itself.

There is no switch/sampler for the error signal $e(t)$

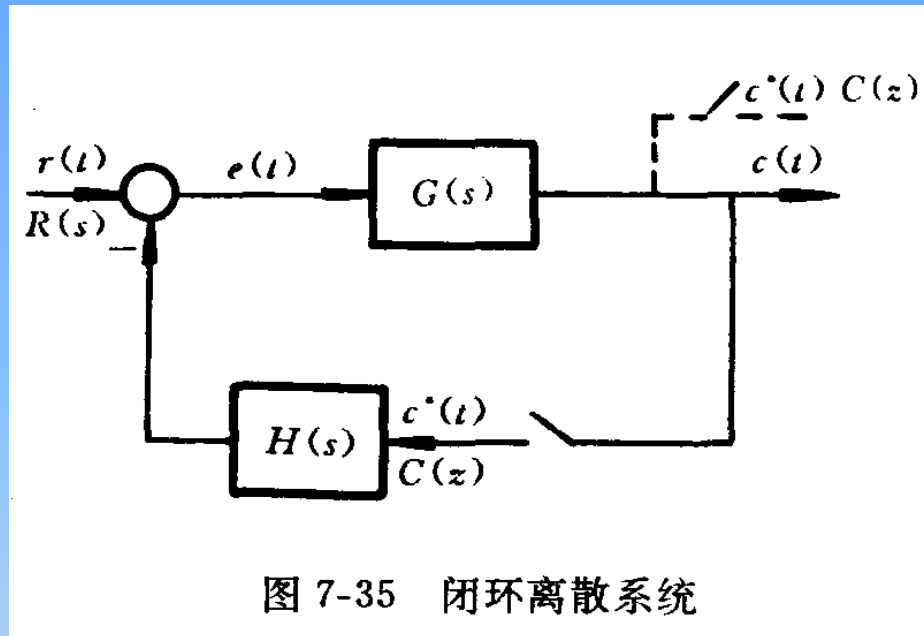


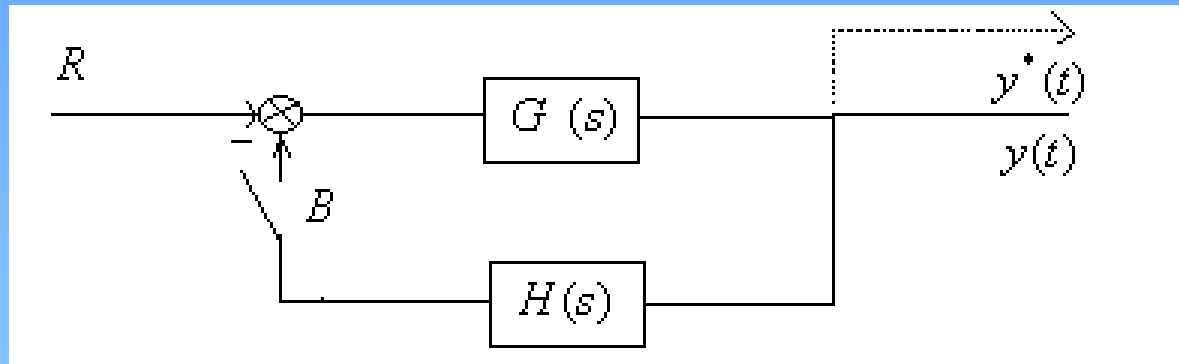
图 7-35 闭环离散系统

$$C(s) = G(s)R(s) - G(s)H(s)C^*(s)$$

$$C(z) = GR(z) - GH(z)C(z) \quad \Rightarrow \quad C(z) = \frac{GR(z)}{1 + GH(z)}$$

Then, for this system, there exists no impulse transfer function.

Example Consider the discrete-time system as shown in the figure, find the z-transform of the output $y(t)$.



Solution:

$$Y(z) = GR(z) - G(z)B(z)$$

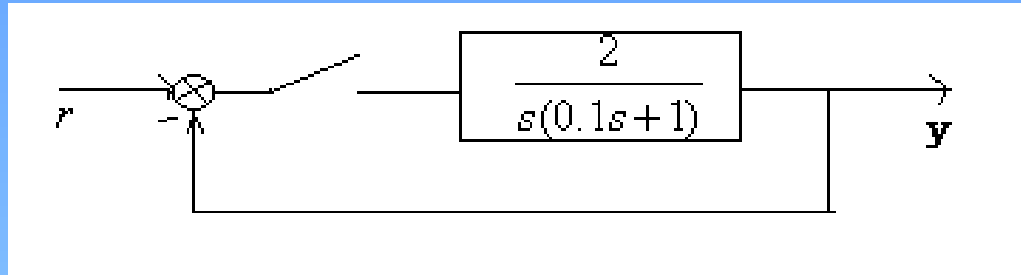
$$B(z) = GHR(z) - GH(z)B(z)$$

$$\therefore B(z) = \frac{GHR(z)}{1 + GH(z)}$$

$$\therefore Y(z) = GR(z) - \frac{G(z)GHR(z)}{1 + GH(z)}$$

There exists no impulse transfer function.

Example Consider the discrete-time system as shown in the figure, for $T=0.1$, find the unit step response of the system.



Solution:
$$G(z) = Z\left[\frac{2}{s(0.1s + 1)}\right] = \frac{2z}{z-1} - \frac{2z}{z - e^{-10T}}$$

$$= \frac{2z - 0.736z}{(z-1)(z-0.368)} = \frac{1.264z}{z^2 - 1.368z + 0.368}$$

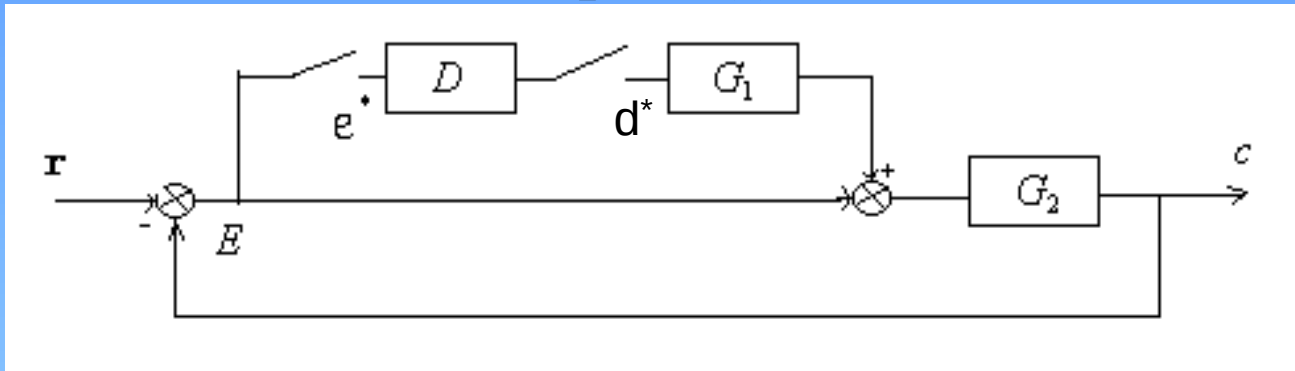
$$\therefore \Phi(z) = \frac{G(z)}{1 + G(z)} = \frac{1.264z}{z^2 - 0.104z + 0.368}$$

$$\therefore Y(z) = \Phi(z)R(z) = \Phi(z)\frac{z}{z-1}$$

$$= 1.264z^{-1} + 1.396z^{-2} + 0.945z^{-3} + 0.849z^{-4} + \dots$$

$$y^*(t) = 1.264\delta(t - 0.1) + 1.396\delta(t - 0.2) + \dots$$

Example Consider the discrete-time system as shown in the figure, find the z-transform of the output $c(t)$.



Solution: There exist both discrete and continuous signals, then employing L-Transform firstly,

$$C(s) = G_2(s)E(s) + G_1G_2D^*E^*$$

$$E(s) = R - C = R - G_2E - G_1G_2D^*E^* \quad \therefore E = \frac{R}{1+G_2} - \frac{G_1G_2}{1+G_2} D^*E^*$$

Discretize $e(t)$, then

$$E^* = \left[\frac{R}{1+G_2} \right]^* - \left[\frac{G_1G_2}{1+G_2} \right]^* D^*E^* \quad \therefore E^* = \frac{\left[\frac{R}{1+G_2} \right]^*}{1 + \left[\frac{G_1G_2}{1+G_2} \right]^* D^*}$$

Take E and E* into

$$C(s) = E(s)G_2(s) + G_1G_2D^*E^*$$

$$C = \frac{G_2R}{1+G_2} - \frac{G_1G_2^2}{1+G_2}D^*E^* + G_1G_2D^*E^*$$

$$= \frac{G_2R}{1+G_2} + \left(-\frac{G_1G_2^2}{1+G_2} + G_1G_2 \right) D^*E^*$$

$$= \frac{G_2R}{1+G_2} + \frac{G_1G_2}{1+G_2} D^*E^*$$

$$\begin{aligned}
 C^*(s) &= \left[\frac{G_2 R}{1+G_2} \right]^* + \left[\frac{G_1 G_2}{1+G_2} \right]^* D^* E^* \\
 &= \frac{\left[\frac{G_2 R}{1+G_2} \right]^* + \left[\frac{G_1 G_2}{1+G_2} \right]^* D^* \left[\left(\frac{G_2 R}{1+G_2} \right)^* + \left[\frac{R}{1+G_2} \right]^* \right]}{1 + \left[\frac{G_1 G_2}{1+G_2} \right]^* D^*}
 \end{aligned}$$

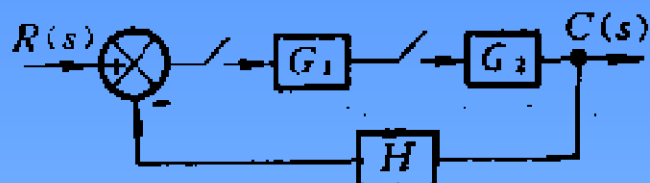
$$\because R = \frac{G_2 R}{1+G_2} + \frac{R}{1+G_2} \quad \therefore R^* = \left[\frac{G_2 R}{1+G_2} \right]^* + \left[\frac{R}{1+G_2} \right]^*$$

$$\therefore C^* = \frac{\left[\frac{G_2 R}{1+G_2} \right]^* + \left[\frac{G_1 G_2}{1+G_2} \right]^* D^* R^*}{1 + \left[\frac{G_1 G_2}{1+G_2} \right]^*}$$

Typical diagram of C.L.discrete-time systems

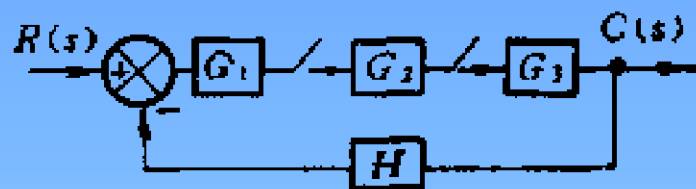
	系 统 方 框 图	$C(z)$
1		$C(z) = \frac{G(z)}{1 + HG(z)} R(z)$
2		$C(z) = \frac{G(z)}{1 + G(z)H(z)} R(z)$
3		$C(z) = \frac{RG(z)}{1 + HG(z)}$
4		$C(z) = \frac{RG_1(z)G_2(s)}{1 + G_1G_2H(z)}$

5



$$C(z) = \frac{G_1(z)G_2(z)}{1 + G_1(z)HG_2(z)} R(z)$$

6



$$C(z) = \frac{G_2(z)G_3(z)RG_1(z)}{1 + G_3(z)G_1G_3H(z)}$$